

Yiqiang Wen

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EDUCATION

China Agricultural University

Beijing, China

Bachelor of Agronomy in Aquaculture

Expected July 2025

Overall GPA: 3.65/4.0

Core Curriculum:	Biochemistry	Cell & Molecular Biology	Genetics	Histology	Botany
	Embryology	Microbiology	Zoology	Physiology	Organic chemistry

Institute of Botany, Chinese Academy of Sciences (IB-CAS)

July 2022 – Dec. 2023

State Key Laboratory of Systematic and Evolutionary Botany

Joint-Training Program, Supervisor: Wenli Chen (Associate Researcher)

RESEARCH PROJECTS

Plant Cell Surface Receptor Function Research

University of Toronto, Mitacs scholarship, Supervisor: Adam Mott (Assistant Professor)

July 2024 – Oct. 2024

Research methods: Genome editing / Genotyping / PCR / Plant cultivation / Stomata image processing

- The Mott lab focuses on the evolution and function of plant cell receptors (LRR-RLKs).
- Screened BRR (broad-range resistance genes)-knockout mutants for responses to pathogen signals and other stresses, using classical methods in addition to high-throughput screening and network analysis, in order to identify the immune function of BRRs.
- Engaged in plant and molecular work and revealed the drought-resistant function through statistical analysis of stomata and water loss data obtained in multiple parallel trials.

Phylogenetic and Systematic Research about Needle Grasses (Poaceae: Pooideae: Stipeae)

IB-CAS, Supervisor: Wenli Chen (Associate Researcher)

Jan. 2024 – Present

Research methods: DNA extraction, amplification, and sequencing / Phylogeny tree / Scanning electron microscopy (SEM) / Chromosome slide-making technique / Principal component analysis(PCA) / academic writing

- Explored the systematic and evolutionary relationships of the genus *Neotrinia* within the needle grass tribe through both fieldwork and laboratory research, and conducted taxonomic revisions.
- This research is based on the analysis of the chloroplast genome, ITS, Cytogenetics (chromosome), and lemma micromorphology.
- Writing a research paper as the **first author** and to be submitted in December 2024.

Eco-Friendly Protein Source: Black Soldier Fly Larvae as a Fishmeal Substitute

China Agricultural University, Supervisor: Songjian Nan (Associate Professor)

Oct. 2022 – Oct. 2024

- Researched the growth characteristics of black soldier fly larvae in mixed bran and tofu residue substrate.
- As the **project leader**, led the team to secure funding, designed and executed feeding trials to collect and assess the growth and health metrics (intestinal enzyme activity, microbiome) of crucian carp.
- Performed data analysis to determine the maximum tolerated and optimal replacement ratios of black soldier fly larvae meal for fishmeal in crucian carp farming.
- Utilized SPSS for data analysis, Prism for graphing, and Visual Basic for coding a data automation tool.

Construction and Mining of Poaceae Species Diversity Database

Chinese Academy of Sciences, Supervisor: Wenli Chen (Associate Researcher)

July 2022 – June 2023

- Conducted over 40 plot surveys of Poaceae plants, collecting biodiversity data such as C3/C4 photosynthetic pathway, DNA sequence, and chromosome ploidy for the Poaceae species diversity database.
- Utilized ArcGIS 10.2 to simulate and calculate environmental variables for each geographical grid.
- Utilized SPSS and R Vegan package to model collected data on biomass, soil moisture, and other ecological variables to identify relationships and patterns within dataset.

INTERNSHIPS

Yantai Institute of Coastal Zone Research, Chinese Academy of Sciences

Aug. 2023 – Dec. 2023

Supervisor: Wenjun Li (Associate Researcher)

- Studied the protein structure and **disease treatment** effectiveness of **phycocyanin** and **phycocyanobilin**.
- Investigated effects of phycocyanin on both pulmonary and gut microbiota in a pulmonary fibrosis model.
- The team explored and revealed the application value of phycocyanin in enhancing immune function and its anti-inflammatory effects on human diseases.
- Designed experiments for constructing plasmids and synthesized phycocyanobilin in *E. coli*.
- Participated in mouse experiments and bioreactor fermentation experiments.

Institute of Botany, Chinese Academy of Sciences (IB-CAS)

Apr. 2022 – June 2022

Supervisor: Wenli Chen (Associate Researcher)

- Responsible for sorting and cataloging plant specimens for the National Herbarium (PE) at IB-CAS.

FIELD EXPEDITION

Multi-Factor Investigation of Heilongjiang Grassland

Chinese Academy of Sciences, Supervisor: Wenli Chen (Associate Researcher)

July 2022 – Aug. 2022

- Participated in a **36-day** field expedition in Heilongjiang, China, across over **3000 kilometers**, conducting herbaceous plant investigation in **41 grassland sample areas**, collecting data of over **1000 plant species**.
- Supplied direct data support for grassland restoration and monitoring of grass-animal balance, and provided the scientific basis for the sustainable development of ecological grassland husbandry in China.

PUBLICATIONS & PATENTS

Rulin Song, **Yiqiang Wen**, Songjian Nan. Exploration of crucian carp (*Carassius auratus*) seedling feeds with the addition of black soldier flies (*Hermetia illucens*) based on the orthogonal test and entropy weighting method. *Journal of Applied Ichthyology*. (Submitted). [In English]

Haoyang Zhang, **Yiqiang Wen**, Junpeng Qu, Songjian Nan. Study on the growth characteristics of the larva of the Black soldier fly in the mixed substrate of bran and bean curd residue. *Hubei Agricultural Science*. 2024, 63 (7): 129-133, 153. [In Chinese]

Yiqiang Wen, Haoyang Zhang, Junyang Li, Songjian Nan. 2002322532515.3: The invention relates to an intelligent assembled larval breeding device of the *Hermetia illucens*. 2023.9.18. Chinese Patent.

Haoyang Zhang, **Yiqiang Wen**, Junyang Li, Songjian Nan. 202322532458.9: The invention relates to an intelligent welfare physiological monitoring chicken coop. 2023.9.18. Chinese Patent.

CONFERENCES ATTENDED & SUMMER PROGRAMS

The 5th Sino-German Block Seminar of IRTG AMAIZE-P, *University of Hohenheim, Germany* 2023

Summer School: Plant Development and Molecular Biology, *Peking University, China* 2023

HONORS & AWARDS

First Prize, *National Undergraduate Competition on Agriculture, Architecture, Environment, and Energy Engineering* 2024

Merit Student, *China Agricultural University* 2024

Literary and Sports Merit Scholarship, *China Agricultural University* 2024

Third Prize in Biochemistry Laboratory Skills, *Biochemistry Innovation & Entrepreneurship Competition* 2023

Second Prize (Top 10%) Academic Scholarship, *China Agricultural University* 2023

Excellent Student Leader, *China Agricultural University* 2022

TECHNICAL SKILLS

Experimental Skills:	bacteria/cell cultivation / plant cultivation / chromosome slide-making technique DNA extraction, amplification, and sequencing / q-PCR / genome editing / genotyping / phylogenetic analyses / phylogeny tree construction protein purification / gel electrophoresis / centrifugation/ mouse experiment scanning electron microscopy (SEM) / fluorescence microscopy
Computer Skills:	SPSS / Solidworks (CAD Modeling) / Visual Basics / Photoshop / webpage building